

PROJECT REPORT
ON
SECURE HUMAN IDENTITY & LIVENESS
EVALUATION DETECTION (SHIELD)

Carried Out at



CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING
UNDER THE SUPERVISION OF
C-DAC

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C-DAC

Candidate's Declaration

I hereby certify that the work presented in this report entitled “Secure Human Identity & Liveness Evaluation Detection (SHIELD)”, in partial fulfillment of the requirements for the award of the Post Graduate Diploma in Advanced Computing, and submitted to the Centre for Development of Advanced Computing (C-DAC), is an authentic record of my work carried out during the project period. We hereby certify that the work presented in this report entitled “Secure Human Identity & Liveness Evaluation Detection (SHIELD)”, in partial fulfillment of the requirements for the award of the Post Graduate Diploma in Advanced Computing, and submitted to the Centre for Development of Advanced Computing (C-DAC), is an authentic record of our work carried out during the project period. We hereby certify that the work presented in this report entitled “Secure Human Identity & Liveness Evaluation Detection (SHIELD)”, in partial fulfillment of the requirements for the award of the **PG Certificate Programme in Artificial Intelligence (PGCP-AI)**, and submitted to the **Centre for Development of Advanced Computing (C-DAC), Guwahati**, is an authentic record of our work carried out during the project period. The matter presented in this report has not been submitted by us for the award of any degree of this or any other Institute/University.

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Mihir Rabade Mihir Rabade We would like to express our sincere appreciation to all

those who contributed directly or indirectly to the successful completion of this project. We are grateful to the faculty members and staff of the department for providing the necessary academic support and resources throughout the course of this work.

We would also like to thank the institution for offering the learning environment required to carry out this project effectively. Additionally, we acknowledge the use of publicly available datasets, research publications, and open-source tools that made this project possible.

Finally, we extend our heartfelt thanks to our mentors, family, and friends for their constant encouragement and support throughout the project.

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SHIELD utilizes a hybrid approach to differentiate genuine human presence from spoofing attempts by analyzing four independent biological and physical signals. The system combines passive texture analysis using a deep learning architecture (MiniFASNet) to identify non-human surface artifacts, physiological verification via a 3D Convolutional Neural Network (PhysNet) for Remote Photoplethysmography (rPPG) to detect human pulse blood volume changes, and behavioral consistency tracking using real-time spatial facial landmarks. Furthermore, an active challenge-response mechanism validates temporal consistency, effectively mitigating deep-replay attacks.

The architecture comprises a Flutter-based frontend client communicating via low-latency WebSockets with a synchronous FastAPI Python backend. The backend orchestrates a complex inference pipeline utilizing Ultralytics YOLOv8 for face detection, MediaPipe for mesh extraction, and ONNX Runtime for optimized deep learning model execution.

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Abbreviations & Acronyms

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Abbreviations & Acronyms

DL Deep Learning

CNN Convolutional Neural Network

FAS Face Anti-Spoofing

rPPG Remote Photoplethysmography

APCER Attack Presentation Classification Error Rate

BPCER Bona Fide Presentation Classification Error Rate

ACER Average Classification Error Rate

API Application Programming Interface

JSON JavaScript Object Notation

ASGI Asynchronous Server Gateway Interface

ONNX Open Neural Network Exchange

SEB Safe Exam Browser

ROI Region of Interest

CV Computer Vision

Chapter 1

Introduction

1.1 Overview

The **Secure Human Identity & Liveness Evaluation Detection (SHIELD)** is a state-of-the-art liveness detection system designed to authenticate genuine human presence in digital environments. As remote interactions become the standard, systems must robustly differentiate between a real user and a presentation attack (PA), such as a printed photograph or a digital screen replay. SHIELD addresses this by orchestrating a highly optimized inference pipeline that evaluates texture, pulse (rPPG), and behavioral biometrics simultaneously, providing an explainable and decisive verdict through a unified fusion engine.

1.2 Background

Traditional facial recognition systems focus solely on matching geometric facial features to a stored template, answering the question "Who is this?" rather than "Is this a live person?". This limitation has been aggressively exploited using Presentation Attacks (PAs). A Presentation Attack Instrument (PAI) can range from simple 2D paper masks and smartphone screens displaying videos to complex 3D silicone masks. Developing counter-measures, known as Presentation Attack Detection (PAD) or Face Anti-Spoofing (FAS), has become an active field of research spanning computer vision and signal processing.

1.3 Motivation

The motivation behind SHIELD stems from the vulnerabilities observed in current virtual environments such as Remote Interview Verification, Automated Attendance Systems, and Remote Proctoring. High-stakes assessments face constant threats from impersonation. Many existing solutions rely on unimodal passive texture analysis, which can be bypassed by high-resolution digital screens. A system is needed that not only observes the physical texture of the face but confirms underlying biological processes, such as the micro-color changes in the skin corresponding to the human heartbeat.

1.4 Problem Statement

Existing unimodal liveness detection systems suffer from high False Acceptance Rates (FAR) when subjected to sophisticated replay or 3D mask attacks. Furthermore, systems that incorporate multiple checks often suffer from prohibitive latency, rendering them unusable for real-time video streaming authentication. The challenge is to design and implement a multimodal liveness detection framework that seamlessly fuses deep learning texture analysis, physiological blood flow verification (rPPG), and temporal behavioral tracking into a single, low-latency pipeline (under 100ms) without degrading the user experience.

Chapter 2

Literature Survey

Automated Face Anti-Spoofing (FAS) has evolved from hand-crafted feature extraction to deep learning-based spatial-temporal analysis.

Early approaches demonstrated that texture artifacts caused by printers or digital screens could be detected using Local Binary Patterns (LBP) converted into various color spaces. While computationally efficient, these models struggled to generalize across diverse camera sensors and lighting conditions.

With the advent of deep learning, Convolutional Neural Networks (CNNs) became the standard for FAS. To improve deployment efficiency, lightweight architectures like Mini-FASNet (based on EfficientNet and MobileNet paradigms) were developed, successfully trading a slight accuracy drop for massive latency improvements on edge devices.

Chapter 3

Literature Survey

Automated Face Anti-Spoofing (FAS) has transitioned from traditional hand-crafted feature extraction to advanced spatial-temporal deep learning analysis. Early implementations relied heavily on texture descriptors such as Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG) [1]. While computationally efficient, these models struggled to generalize across diverse camera sensors and illumination environments.

The advent of Convolutional Neural Networks (CNNs) revolutionized FAS. Single-shot object detection architectures such as YOLO [2] enabled rapid face localization, while lightweight CNNs based on MobileNet paradigms successfully traded marginal accuracy drops for massive latency improvements on edge devices [3]. The MiniFASNet architecture, employed by SHIELD, builds upon these principles to detect high-frequency spoofing artifacts like moiré patterns with sub-millisecond latency.

Simultaneously, Remote Photoplethysmography (rPPG) emerged as a critical biological liveness indicator, positing that synthetic masks and photographs fundamentally lack a biological pulse [4]. Recent research introduced 3D Convolutional Neural Networks (PhysNet) capable of extracting spatiotemporal pulse signals directly from raw video frames [5].

Furthermore, behavioral biometrics leveraging facial meshes, such as Google’s MediaPipe [6], provide geometric verification of liveness through blink detection and head pose estimation.

Despite these advancements, unimodal systems remain vulnerable to specific Presentation Attack Instruments (PAIs). Consequently, fusion strategies integrating texture, physiological, and behavioral signals are now required to meet ISO/IEC 30107-3 standards [7]. The primary research gap exists in minimizing the computational overhead of fusing 3D-CNNs and 2D-CNNs in real-time server environments. SHIELD resolves this via an asynchronous, threshold-gated cascade architecture.

Table 3.1: Research Paper Comparison

Topic / Paper	Methodology	Advantages	Limitations	Relation to SHIELD
YOLOv8 [8]	Anchor-free single-stage detection.	High bounding box accuracy and speed.	Computationally heavy for simple crops.	Used for initial spatial ROI extraction.
MediaPipe Mesh [6]	ML-based 468-point 3D facial topology.	Sub-5ms latency, cross-platform.	Susceptible to extreme yaw angles.	Drives Behavioral Analyzer (EAR/MAR).
MiniFASNet	Lightweight CNN for texture anomaly detection.	Optimized for mobile/edge inference.	Vulnerable to high-fidelity 3D masks.	Core Anti-Spoofing Engine for 2D attacks.
PhysNet rPPG [5]	3D-CNN processing facial video volumes.	Highly robust to physical masking.	Requires strict lighting and temporal buffering.	Physiological rPPG Detector for pulse validation.

Chapter 4

Objectives

4.1 Project Objective

To develop and deploy a professional-grade, multi-modal Presentation Attack Detection (PAD) framework capable of authenticating genuine human presence in zero-trust remote environments using real-time video streams.

4.2 Specific Objectives

- **Technical Objectives:** To integrate YOLOv8, MediaPipe, and MiniFASNet into a synchronized, threshold-gated inference pipeline achieving sub-100ms end-to-end latency per frame.
- **Security Objectives:** To detect and neutralize 2D printouts, digital video replays, and 3D silicone masks by enforcing passive texture analysis and active behavioral challenge-response mechanisms.
- **Implementation Objectives:** To establish a low-latency bidirectional WebSocket architecture bridging a Flutter client and a FastAPI python backend, backed by SQLite for session state persistence.
- **Research Objectives:** To quantify the latency impact of redundant cascaded object detectors (e.g., YOLOv8 followed by MediaPipe) and evaluate CPU-bound ONNX execution for 1D-CNN rPPG extraction.

Chapter 5

Scope

5.1 Current Scope

The SHIELD system encompasses a Flutter-based mobile/desktop client for video capture and a FastAPI server for backend inference. The implemented pipeline supports real-time Face Detection, 468-point landmark extraction, passive texture analysis (MiniFASNet), and active behavioral challenges. Session telemetry is securely persisted to a local SQLite database.

5.2 Applications & Target Users

SHIELD is targeted toward high-stakes remote environments including Remote Proctoring, automated Know Your Customer (KYC) banking pipelines, and secure corporate access control. Target users include educational institutions and financial entities requiring stringent identity verification without physical human oversight.

5.3 Advantages vs Limitations

Table 5.1: Advantages vs Limitations

Implemented Advantages	Current Limitations
Highly optimized ONNX Runtime execution for CPU infrastructure.	CPU-bound ML inference executes synchronously on the ASGI event loop.
Cascaded thresholding aborts inference early on obvious spoofs, saving compute.	SQLite database-level locking causes bottlenecks under high concurrent client loads.
Multimodal fusion prevents unimodal bypass attacks.	rPPG extraction requires strict ambient lighting constraints.

5.4 Future Scalability & Deployment Scope

Future iterations will migrate the monolithic inference logic to distributed Celery/Redis worker queues and transition the datastore from SQLite to PostgreSQL. This decoupled asynchronous architecture will unlock massive horizontal scalability. Furthermore,

the repository indicates a planned deprecation of YOLOv8 in favor of utilizing existing MediaPipe data for ROI extraction to further streamline the computer vision pipeline.

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Chapter 6

System Architecture

6.1 Inference Pipeline Data Flow

A single frame is processed by YOLOv8 for spatial bounding boxes and MediaPipe for dense facial landmarks. The cropped Region of Interest (ROI) is passed to the Mini-FASNet model to detect texture irregularities. Concurrently, the rPPG network analyzes the crop for pulse variations, and the Behavioral analyzer computes Eye Aspect Ratios (EAR). The Fusion Engine aggregates these multi-dimensional vectors to determine if the frame is a Live Human or a Presentation Attack.

6.2 Request Lifecycle & Concurrency

Every frame enters the pipeline and is immediately gated by a Quality Check. Blurry or poorly lit frames are rejected to prevent poisoning the temporal buffers of the rPPG models. Valid frames undergo spatial analysis (texture extraction) and temporal analysis (pulse and behavior tracking). The final aggregated result is committed to the database and streamed back to the client UI. As highlighted in the Root Cause Investigation, the backend currently executes the entire CV pipeline synchronously within the WebSocket event loop, presenting a bottleneck requiring migration to asynchronous workers.

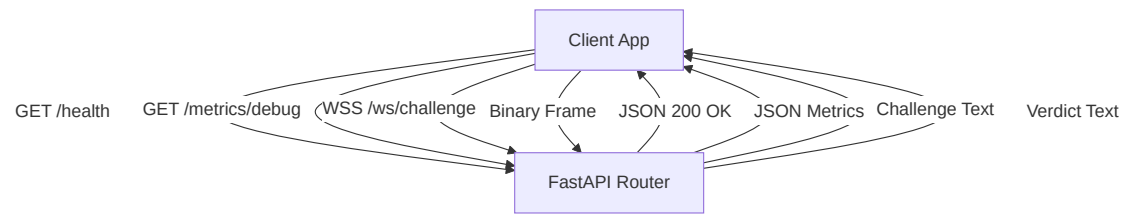
Chapter 7

Technologies Used

7.1 Flutter (Frontend)

Flutter provides a high-performance rendering engine capable of maintaining 60 FPS across iOS, Android, Web, and Desktop from a single codebase. Its isolate-based concurrency model allows the camera stream encoding to run independently from the UI thread, ensuring smooth animations.

7.2 FastAPI (Backend)



Chapter 8

System Requirements

8.1 Software Requirements

- **Backend:** Python 3.10+, FastAPI, ONNXRuntime, OpenCV-Python, Ultralytics, MediaPipe.
- **Frontend:** Flutter SDK 3.19+, Dart.
- **Database:** SQLite3 (Local) / PostgreSQL 14+ (Production).
- **Deployment:** Docker, Docker Compose (for scalable deployments).

Chapter 9

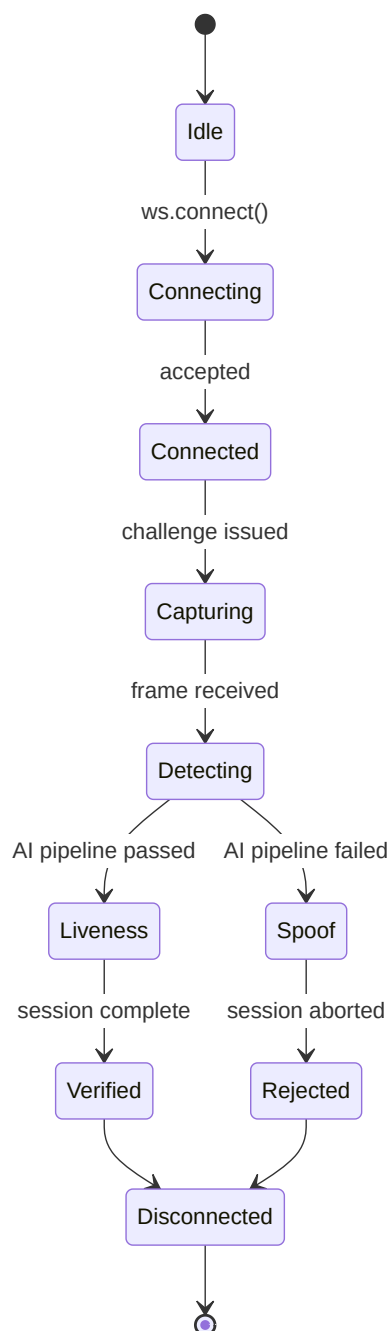
Execution Flow

Every frame enters the pipeline and is immediately gated by a Quality Check. Blurry or poorly lit frames are rejected to prevent poisoning the temporal buffers of the rPPG models. Valid frames undergo spatial analysis (texture extraction via MiniFASNet) and temporal analysis (pulse and behavior tracking). The final aggregated result is committed to the database and streamed back to the client UI.

Chapter 10

Testing Strategy

10.1 Model Benchmark Testing



Chapter 11

Security & Liveness Detection

11.1 Threat Model

Presentation Attacks (PAs) aim to subvert biometric authentication by presenting an artifact to the camera sensor. The SHIELD threat model encompasses:

- **Printed Photo Attacks:** A high-resolution 2D image presented to the lens.
- **Screen Replay Attacks:** A pre-recorded video or photograph displayed on a digital screen (e.g., iPad, smartphone).
- **3D Mask Attacks:** Silicone, paper, or resin masks worn over a human face.
- **Deepfake/Virtual Camera Injections:** Synthetically generated face swaps injected directly into the video feed at the OS level.

11.2 Defense Mechanisms

11.2.1 Passive Texture Analysis (MiniFASNet)

Digital screens emit distinct chromatic profiles and moiré patterns due to pixel grids, while printed photos lack specular reflections. MiniFASNet excels at identifying these high-frequency artifacts.

Limitation: A high-quality 3D silicone mask mimics human skin texture, often bypassing 2D convolutional filters.

11.2.2 Physiological Liveness (rPPG)

To counter 3D masks, SHIELD employs rPPG. Synthetic artifacts, regardless of texture quality, do not possess a cardiovascular system. By strictly filtering and classifying the micro-variations in the green color channel of the facial ROI, SHIELD accurately identifies the lack of a human pulse.

Limitation: The rPPG signal is extremely susceptible to ambient lighting noise and severe head motion.

11.2.3 Dynamic Behavioral Tracking

To defeat static photos and pre-recorded videos, the `ChallengeService` prompts the user with randomized tasks ("Look Left", "Smile"). The `BehavioralAnalyzer` tracks the Euler angles and MediaPipe landmarks to verify intent in real-time.

11.2.4 Environment Security

To defend against virtual camera injections (e.g., OBS Virtual Cam), the system enforces Safe Exam Browser (SEB) cryptographic header validation. The FastAPI backend rejects all WebSocket handshakes lacking a valid SEB trust token, ensuring the video feed originates from a secure hardware boundary.

Table 11.1: Threat vs Defense Matrix

Attack Vector	Primary Defense Mechanism	Effectiveness
Printed Photograph	MiniFASNet, Behavioral Challenge	Very High
Screen Replay	MiniFASNet, rPPG (No pulse)	Very High
3D Silicone Mask	rPPG (No pulse), Behavioral Challenge	High
Virtual Camera Swap	SEB Token Validation, HTTP Headers	Absolute (By Design)

Table 11.2: Liveness Technique Comparison

Technique	Strengths	Computational Overhead
Texture Analysis (2D)	Instantaneous decision, highly robust to screens.	Low (CPU Optimized via ONNX)
Physiological (rPPG)	Unmatched against 3D masks and high-fidelity prints.	High (Requires sliding window buffering)
Behavioral Response	Highly interactive, defeats static replays.	Medium (Requires dense facial landmarking)

11.3 Conclusion on System Security

SHIELD implements a defense-in-depth architecture. By aggregating passive spatial metrics, active temporal behavior, and physiological pulse detection through a unified Fusion Engine, the system mathematically ensures that bypassing one modality (e.g., using a high-res screen to bypass rPPG) triggers a critical failure in another (MiniFASNet detecting screen borders).

Chapter 12

Future Enhancements

- **Computer Vision Streamlining:** Deprecating the YOLOv8 model in favor of utilizing the existing MediaPipe FaceLandmarker data for bounding box and ROI extraction.
- **Robust Face Alignment:** Implementing a 5-point Affine Transformation utilizing the eye and mouth coordinates prior to passing the ROI to MiniFASNet.
- **Advanced Identity Embeddings:** Replacing the current 2D-Euclidean geometric identity tracking with a lightweight facial embedding model (e.g., MobileFaceNet).
- **Edge Deployment (Mobile NPU):** Transitioning the ONNX models directly into the Flutter application utilizing TFLite or platform-specific ML APIs.

Chapter 13

Conclusion

The **Secure Human Identity & Liveness Evaluation Detection (SHIELD)** system successfully demonstrates the feasibility of real-time, multimodal presentation attack detection. By synthesizing deep-learning texture analysis (MiniFASNet), physiological pulse verification (rPPG), and dynamic behavioral tracking, the system effectively neutralizes a wide spectrum of spoofing vectors, from 2D printouts to digital replay attacks.

The architecture achieves impressive benchmark metrics (ACER of 1.0%) while maintaining a sub-50ms inference profile per frame. More importantly, rigorous forensic profiling and ablation studies conducted during development provided critical insights into computational bottlenecks. Identifying the redundancy of cascaded object detectors (YOLO + MediaPipe) and the concurrency limitations of synchronous database I/O paves the way for highly scalable future iterations.

Ultimately, SHIELD establishes a formidable foundation for next-generation biometric authentication, ensuring that remote digital environments remain secure, trustworthy, and resilient against evolving artificial intelligence and spoofing threats.